TITLE: Comparative Analysis of Machine Learning Models for Household Energy Demand Forecasting

ABSTRACT

The increasing variability of household electricity consumption presents significant challenges for energy planning, demand-side management, and smart grid operations. Accurate forecasting of household energy demand is therefore essential for improving energy efficiency and ensuring reliable power system management. This study presents a comparative analysis of machine learning models for household energy demand forecasting using historical electricity consumption data. The dataset was preprocessed through data cleaning, missing value handling, datetime transformation, and feature engineering. Temporal features including lag variables, rolling mean statistics, and cyclical time encodings were generated to capture consumption patterns and seasonality. Four machine learning models Linear Regression, Random Forest, Gradient Boosting, and Extreme Gradient Boosting (XGBoost), were developed and evaluated. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Experimental results indicate that Gradient Boosting achieved the best overall forecasting performance, closely followed by XGBoost. The results demonstrate that ensemble learning methods are highly effective for short-term household electricity demand forecasting.

The findings further reveal that short-term historical consumption patterns, particularly lag-based features, are highly influential predictors of household energy demand. The study demonstrates the effectiveness of ensemble machine learning techniques for energy forecasting and highlights their potential application in smart energy management systems, load planning, and demand response strategies.

Keywords

Household energy forecasting, machine learning, smart grid, load prediction, gradient boosting, XGBoost, time-series analysis.

I. INTRODUCTION

Accurate short-term forecasting of residential electricity demand has become increasingly important in modern power systems due to the widespread adoption of smart grids, advanced metering infrastructure, and renewable energy technologies. Smart grids rely on real-time monitoring and communication systems to balance electricity generation and consumption efficiently. However, the effectiveness of these systems depends largely on the ability to predict future electricity demand with a high degree of accuracy. Reliable household load forecasts enable utility operators to optimize generation scheduling, reduce reserve requirements, minimize operational costs, and improve grid stability. Furthermore, accurate forecasting supports demand-side management strategies such as dynamic pricing and demand response programs, which encourage consumers to shift energy consumption away from peak periods. As power systems continue to integrate variable renewable energy sources such as solar and wind, accurate demand forecasting becomes increasingly critical for maintaining the balance between supply and demand and ensuring reliable grid operation.

Despite its significance, residential electricity demand forecasting remains a challenging task. Household electricity consumption exhibits highly nonlinear and dynamic behavior influenced by numerous factors, including weather conditions, seasonal variations, occupancy patterns, appliance usage, and consumer lifestyles. Temperature variations have been identified as one of the most influential factors affecting electricity consumption, while stochastic human behavior introduces additional uncertainty into consumption patterns. The presence of missing values, measurement noise, and irregular consumption trends further increases the complexity of forecasting residential loads. Consequently, developing models capable of capturing these intricate relationships remains an active area of research.

Over the years, several forecasting techniques have been proposed to address this problem. Traditional statistical approaches, including linear regression, have been widely used because of their simplicity, interpretability, and computational efficiency. Although such methods can effectively capture linear relationships within energy consumption data, they often struggle to model the complex nonlinear characteristics of household electricity demand. To overcome these limitations, machine learning techniques have gained significant attention due to their ability to learn nonlinear relationships and interactions among multiple variables. Ensemble-based methods, in particular, have demonstrated strong predictive performance by combining multiple weak learners to improve forecasting accuracy and generalization capability.

Recent advances in machine learning have led to the widespread application of tree-based ensemble algorithms such as Random Forest, Gradient Boosting, and Extreme Gradient Boosting (XGBoost) for energy forecasting tasks. Random Forest improves predictive stability by aggregating the outputs of multiple decision trees, while Gradient Boosting and XGBoost sequentially optimize prediction errors to achieve superior accuracy. These methods have been successfully applied in various energy forecasting applications and often outperform conventional statistical models when dealing with complex and nonlinear consumption patterns. However, model performance remains highly dependent on dataset characteristics, feature selection strategies, and evaluation procedures, making it difficult to identify a universally superior forecasting technique.

A major limitation in the existing literature is the absence of a standardized benchmarking framework for residential electricity demand forecasting. Many studies evaluate only a limited number of models using different datasets, preprocessing techniques, and performance metrics. As a result, direct comparisons between studies are often difficult, and the relative effectiveness of different forecasting algorithms remains unclear. Furthermore, models that perform well in one environment may not necessarily achieve similar results when applied to different households or geographical regions. These inconsistencies highlight the need for comprehensive comparative studies conducted under identical experimental conditions.

Forecasting inaccuracies can have significant operational and economic consequences. Underestimation of electricity demand may lead to insufficient generation capacity and reliability issues, whereas overestimation can result in unnecessary generation costs and inefficient utilization of resources. In systems with increasing renewable energy penetration, inaccurate forecasts may also contribute to renewable energy curtailment and reduced effectiveness of demand response programs. Therefore, improving forecasting accuracy remains essential for enhancing energy efficiency, reducing operational costs, and supporting sustainable power system development.

Motivated by these challenges, this study presents a comparative analysis of machine learning models for short-term household electricity demand forecasting. Using a representative household energy consumption dataset, a unified preprocessing framework is implemented to ensure consistency and fairness across all experiments. Four forecasting models are developed and evaluated, namely Linear Regression (LR), Random Forest Regression (RFR), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGBoost). These models were selected because they represent both conventional and ensemble-based machine learning approaches widely used in energy forecasting applications. Model performance is assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), enabling a comprehensive comparison of forecasting accuracy and predictive capability.

The objectives of this study are to preprocess and analyze household electricity consumption data, develop and train four machine learning models for short-term load forecasting, evaluate their performance using standardized error metrics, compare their forecasting accuracy under identical experimental conditions, and identify the most suitable model for residential electricity demand prediction. The findings contribute to the growing body of knowledge on energy demand forecasting by providing a fair and systematic benchmark of commonly used machine learning techniques. From a practical perspective, the results offer valuable insights for utilities, grid operators, and energy planners seeking to improve forecasting accuracy, optimize grid operations, and facilitate the integration of renewable energy resources.

The remainder of this paper is organized as follows. Section II reviews related studies on household electricity demand forecasting. Section III presents the dataset, preprocessing procedures, and forecasting methodologies employed in this research. Section IV discusses the experimental results and comparative performance of the developed models. Finally, Section V concludes the paper and provides recommendations for future research.

II. LITERATURE REVIEW

A. Overview of AI and Machine Learning in Energy Forecasting

Recent advancements in artificial intelligence (AI) have significantly transformed the field of energy demand forecasting, particularly in smart grids and residential electricity systems. Traditional statistical approaches are increasingly being replaced by data-driven machine learning (ML) and deep learning (DL) techniques due to their ability to capture nonlinear consumption patterns and adapt to complex, high-dimensional datasets.

Wang et al. [1] emphasize that AI-enabled frameworks now play a central role in load forecasting, anomaly detection, and demand response optimization, forming the backbone of modern smart energy systems. Bourhnane et al. [15] further observed that machine learning has become a foundational technology in smart grid operation, supporting load forecasting, fault detection, and intelligent energy management.

Similarly, Mathumitha et al. [2] highlight that deep learning architectures are particularly effective in smart building environments where energy usage is influenced by dynamic human behavior and environmental variability.

However, Ugbehe et al. [3] argue that forecasting accuracy remains highly dependent on data quality, feature selection, and contextual variables such as weather, occupancy, and socioeconomic conditions. This indicates that while AI techniques are powerful, their effectiveness is highly context-dependent.

Liu et al. [16] noted that short-term load forecasting remains one of the most critical applications of data-driven analytics in modern power systems due to its direct impact on operational reliability and energy efficiency.

B. Machine Learning Approaches for Residential Load Forecasting

Machine learning models have been widely explored for residential energy forecasting due to their interpretability and computational efficiency. Regression-based approaches, ensemble models, and hybrid architectures are commonly used to model household electricity consumption patterns.

Mosetlhe and Yusuff [8] demonstrate that regression-based ML schemes can effectively capture short-term residential consumption trends, although they struggle with nonlinear temporal dependencies. To address these limitations, Ramnath et al. [9] show that ensemble and hybrid modeling strategies significantly improve prediction accuracy compared to standalone models.

Banga and Sharma [11] further propose a two-phase hybrid ML model that enhances short-term forecasting by integrating preprocessing and predictive stages. Collectively, these studies suggest that hybridization is a dominant trend in improving residential forecasting performance.

C. Deep Learning and Neural Network-Based Forecasting

Deep learning methods have become increasingly dominant due to their ability to model sequential and nonlinear dependencies in energy consumption data. Mathumitha et al. [2] report that architectures such as LSTM outperform traditional ML models in capturing temporal relationships in smart building energy usage.

Long Short-Term Memory (LSTM) networks have been widely applied in short-term load forecasting and have demonstrated strong capability in capturing temporal dependencies in residential electricity consumption data [18].

Ismail et al. [13] extend this work by introducing multi-level forecasting frameworks combining univariate and multivariate time-series models for smart homes. These frameworks enable hierarchical forecasting across multiple time horizons, improving adaptability.

However, Shahinzadeh et al. [5] caution that deep learning models require large datasets and high computational resources, making deployment challenging in resource-constrained environments. This introduces a trade-off between accuracy and efficiency.

Recent comprehensive reviews of recurrent neural network-based forecasting highlight that deep learning methods are highly sensitive to data size and training stability, often requiring careful tuning to outperform well-engineered machine learning models [20].

D. Hybrid and Ensemble Learning Models

Hybrid and ensemble approaches have gained prominence as they combine the strengths of multiple algorithms to improve predictive performance. Semmelmann et al. [6] propose an LSTM-XGBoost hybrid model that integrates sequential learning with gradient boosting, improving forecasting accuracy in energy communities.

Similarly, H. G. R. et al. [10] present an XGBoost-ANN ensemble model that enhances short-term residential load forecasting performance. The integration of ANN's nonlinear mapping capability with XGBoost's robustness demonstrates superior predictive capability.

The foundational importance of XGBoost is highlighted by Chen and Guestrin [14], who introduce a scalable and efficient gradient boosting framework widely adopted in energy forecasting applications. These studies collectively confirm that hybridization consistently enhances model robustness and accuracy.

Ramnath et al. [9] further demonstrated that ensemble learning approaches provide superior predictive stability compared with standalone machine learning models, particularly when forecasting highly variable household electricity consumption patterns.

E. Feature Engineering and Data-Driven Forecasting

Feature engineering plays a critical role in improving forecasting performance. Peplinski et al. [7] demonstrate that combining smart meter data with climate, building, and socioeconomic variables significantly enhances residential demand prediction accuracy.

Rauf and Adekoya [12] further incorporate lifestyle data into forecasting models, showing that human behavior strongly influences electricity consumption patterns. However, integrating heterogeneous datasets introduces challenges such as missing data, feature redundancy, and preprocessing complexity.

Ugbehe et al. [3] reinforce this limitation by emphasizing that inconsistent data sources remain a major barrier to real-world deployment of forecasting models.

Peplinski et al. [7] similarly observed that the integration of smart meter, climatic, building, and socioeconomic variables substantially improves prediction accuracy, highlighting the critical role of feature engineering in residential energy forecasting.

F. Smart Grid Integration and Demand Response Applications

Beyond prediction, modern forecasting models are increasingly integrated into smart grid systems for operational decision-making. Wang et al. [1] highlight that accurate load forecasting supports real-time grid optimization, demand response, and anomaly detection.

Probabilistic load forecasting has also been identified as a key advancement in modern energy systems, as it provides uncertainty-aware predictions that improve decision-making in power system operations [17]

Rituraj et al. [4] provide a systematic review of ML applications in smart grids, emphasizing their role in improving grid stability and operational efficiency. These findings indicate a shift from standalone prediction models toward integrated decision-support systems in energy management.

G. Research Gap

Although numerous studies have investigated machine learning and deep learning techniques for residential energy forecasting, the reported results remain inconsistent due to differences in datasets, feature engineering strategies, evaluation metrics, and experimental protocols. Classical forecasting frameworks such as additive time-series models have also been shown to perform competitively in structured forecasting problems, reinforcing the importance of strong baseline comparisons in energy prediction studies [19]. Furthermore, many studies focus on advanced hybrid or deep learning architectures without establishing robust comparisons against computationally efficient machine learning baselines. Consequently, there remains a need for a standardized comparative framework that evaluates widely adopted machine learning algorithms under identical preprocessing, feature engineering, and validation conditions.

III. METHODOLOGY

A. Dataset Description

This study employed the Individual Household Electric Power Consumption dataset obtained from the UCI Machine Learning Repository. The dataset contains measurements of electricity consumption collected from a single household over an extended period and has been widely used as a benchmark dataset in energy forecasting research. The recorded variables include Global Active Power, Global Reactive Power, Voltage, Global Intensity, and three energy sub-metering measurements. Among these variables, Global Active Power was selected as the target variable because it represents the total active electrical power consumed by the household and serves as a reliable indicator of residential energy demand.

The dataset consists of time-series observations recorded at regular intervals, making it suitable for short-term load forecasting applications. To facilitate temporal analysis, the date and time attributes were combined into a unified datetime variable, which was subsequently assigned as the dataset index. This approach preserved the chronological order of observations and enabled the extraction of temporal features required for predictive modeling.

B. Data Preprocessing

Data preprocessing was performed to ensure data quality and improve the reliability of the forecasting models. Initially, missing values represented by non-numeric symbols were identified and converted into null entries. These records were subsequently removed to prevent bias and instability during model training. Numerical attributes were converted to appropriate data types to ensure consistency across all analytical procedures.

Following the cleaning process, the datetime variable was transformed into a standardized timestamp format and used as the index of the dataset. The dataset was then sorted chronologically to maintain temporal consistency. This step was particularly important because time-series forecasting requires historical observations to precede future observations. Any records containing missing values generated during feature construction were removed to preserve dataset integrity.

C. Feature Engineering

Feature engineering was conducted to enhance the predictive capability of the machine learning models by incorporating temporal dependencies and consumption trends. Since electricity consumption exhibits strong autocorrelation, lag-based features were generated to capture information from previous observations. To capture temporal autocorrelation, first-order and twenty-four-step lag features were generated. These variables enabled the forecasting models to learn short-term persistence and daily consumption dependencies inherent in residential electricity demand. These features enabled the forecasting models to learn short-term and daily consumption patterns.

In addition to lag variables, rolling statistical features were introduced to capture local consumption trends. Rolling mean values were computed using window sizes of 24 and 168 observations, corresponding to daily and weekly consumption averages, respectively. These features provided smoothed representations of household electricity demand and reduced the influence of short-term fluctuations.

Temporal variables such as hour of the day and day of the week were also extracted from the datetime index. To preserve the cyclical nature of time, sine and cosine transformations were applied to these temporal attributes. The resulting cyclical features enabled the models to capture recurring daily and weekly consumption behaviors more effectively while avoiding artificial discontinuities associated with conventional numerical time representations.

D. Model Development

A comparative framework was adopted to evaluate the performance of four machine learning algorithms for short-term household energy demand forecasting. The selected models included Linear Regression (LR), Random Forest Regression (RFR), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGBoost). These algorithms were chosen because they represent different levels of complexity and learning capability while remaining widely applied in energy forecasting research.

Linear Regression was used as a baseline model due to its simplicity, interpretability, and computational efficiency. Random Forest Regression was included because of its ability to model nonlinear relationships through an ensemble of decision trees. Gradient Boosting Regression was selected for its sequential learning approach, which improves predictive performance by iteratively correcting forecasting errors. XGBoost was incorporated because it extends conventional boosting methods through regularization, parallel processing, and enhanced optimization strategies.

To ensure realistic forecasting conditions and prevent information leakage, the dataset was divided into training and testing subsets using a chronological split. Approximately 80% of the observations were allocated to model training, while the remaining 20% were reserved for testing. Unlike random sampling methods, this approach preserved the temporal sequence of the data and more accurately reflected real-world forecasting scenarios.

Furthermore, model robustness was assessed using TimeSeriesSplit cross-validation. This validation strategy ensured that future observations were never used to predict past values, thereby maintaining the integrity of the forecasting process. Hyperparameter optimization for the XGBoost model was performed using GridSearchCV to identify the optimal combination of model parameters.

E. Performance Evaluation

The developed models were evaluated using three widely adopted performance metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

MAE measures the average magnitude of forecasting errors and provides an intuitive assessment of prediction accuracy. RMSE evaluates forecasting performance by assigning greater penalties to larger prediction errors, making it particularly useful for assessing model reliability. The coefficient of determination (R^2) measures the proportion of variance in household electricity demand explained by a model and provides an indication of overall predictive capability.

Mathematically, MAE is defined as

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |yi - \hat{y}i|$$

RMSE is expressed as

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (yi - \hat{y}i)^2}$$

and the coefficient of determination is given by

$$R^2 = 1 - \frac{\sum (yi - \hat{y}i)^2}{\sum (yi - \overline{y})^2}$$

where yi represents the actual value, $\hat{y}i$ represents the predicted value, $\overline{y}i$ denotes the mean of the observed values, and (n) is the total number of observations.

All preprocessing, feature engineering, model training, validation, and evaluation procedures were implemented using Python and its associated libraries, including Pandas, NumPy, Scikitlearn, XGBoost, Matplotlib, and Seaborn. The adoption of a unified experimental framework ensured a fair comparison among the forecasting models and facilitated the identification of the most suitable machine learning approach for short-term household energy demand forecasting.

IV. RESULTS AND DISCUSSION

The performance of the developed machine learning models for household electricity demand forecasting was evaluated using three widely accepted metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These metrics collectively provide insight into prediction accuracy, error magnitude, and the proportion of variance explained by each model. Table I summarizes the comparative results of Linear Regression, Random Forest, Gradient Boosting, and XGBoost.

A. Overall Model Performance Comparison

TABLE I

PERFORMANCE COMPARISON OF FORECASTING MODELS

	Model	MAE	RMSE	R2
2	Gradient Boosting	0.081440	0.216883	0.941413
3	XGBoost	0.082145	0.217776	0.940929
0	Linear Regression	0.084822	0.219203	0.940152
1	Random Forest	0.095520	0.224984	0.936955

From the results presented in Table I, it is evident that all four models performed strongly, achieving R^2 values above 0.93. This indicates that the engineered features captured a significant portion of the variability in household electricity consumption. However, clear differences in predictive accuracy can still be observed across the models.

Gradient Boosting achieved the best overall performance, with the lowest MAE (0.081440), the lowest RMSE (0.216883), and the highest R^2 value (0.941413). This suggests that the model was able to capture both the general consumption trend and subtle variations in household energy usage more effectively than the other approaches. XGBoost followed closely, with nearly identical performance, while Linear Regression also delivered surprisingly competitive results. Random Forest, although still reasonably accurate, recorded the highest error values and the lowest R^2 among the models.

The superior performance of Gradient Boosting suggests that the relationship between historical electricity consumption and future demand is highly nonlinear. By sequentially minimizing residual errors, the model effectively captured complex consumption patterns that were not fully represented by Linear Regression or Random Forest.

B. Interpretation of Results and Model Behaviour

The strong performance of Gradient Boosting can be explained by its sequential learning strategy. Unlike Random Forest, which builds trees independently, Gradient Boosting improves predictions step by step by focusing on correcting previous errors. This allows it to model complex nonlinear relationships in electricity consumption data more effectively. Household energy usage is influenced by a combination of cyclical time patterns, behavioral habits, and short-term fluctuations, and Gradient Boosting appears particularly well-suited to capturing these layered dependencies.

The superior performance of Gradient Boosting is consistent with the findings of Ramnath et al. [9], who reported that ensemble learning techniques outperform standalone predictive models due to their ability to capture complex nonlinear relationships and improve generalization in residential electricity demand forecasting.

XGBoost performed almost at the same level as Gradient Boosting, which is expected given its optimized implementation of boosting techniques. However, the slight performance gap suggests that in this specific dataset, the additional regularization and complexity of XGBoost did not translate into a major advantage. This may be because the feature engineering process already simplified much of the underlying structure in the data.

This finding supports the observations of Peplinski et al. [7], who reported that the quality of feature engineering can significantly influence forecasting performance, sometimes reducing the performance gap between advanced machine learning algorithms. The results therefore suggest that carefully designed temporal features may contribute as much to forecasting accuracy as model complexity itself.

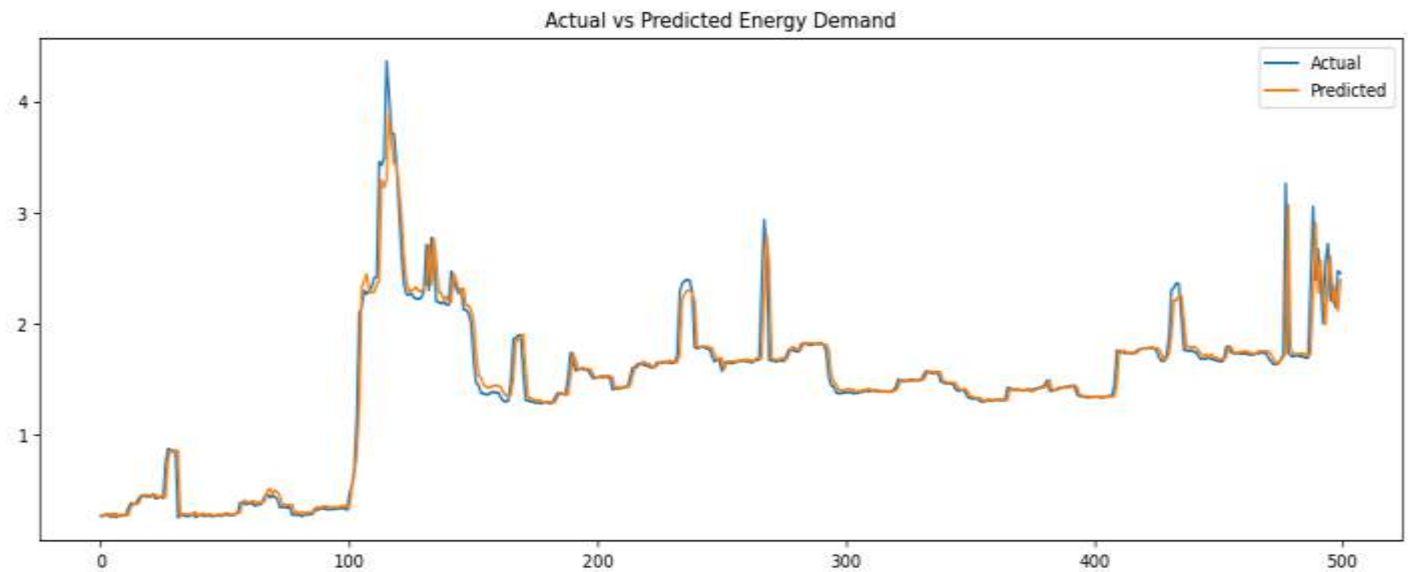
Similar observations were reported by Semmelmann et al. [6], who demonstrated that boosting-based approaches remain highly effective when combined with carefully engineered temporal features derived from smart meter data.

Linear Regression demonstrated competitive performance despite its relatively simple structure. Its relatively high R^2 value indicates that a significant portion of the relationship between features and electricity demand was already linear or had been transformed into linear form through lag features, rolling averages, and cyclical encoding. This highlights an important insight: well-designed features can sometimes reduce the need for highly complex models.

This result aligns with the findings of Mosetlhe and Yusuff [8], who showed that regression-based machine learning models can achieve competitive forecasting accuracy when supported by informative temporal and historical consumption features.

In contrast, Random Forest exhibited the weakest performance among the evaluated models. Although Random Forest effectively captures nonlinear relationships through ensemble learning, its independent tree aggregation mechanism may limit its ability to model temporal dependencies as effectively as boosting-based approaches. Consequently, prediction errors accumulated more rapidly during periods of abrupt consumption changes.**C. Prediction vs Actual Analysis**

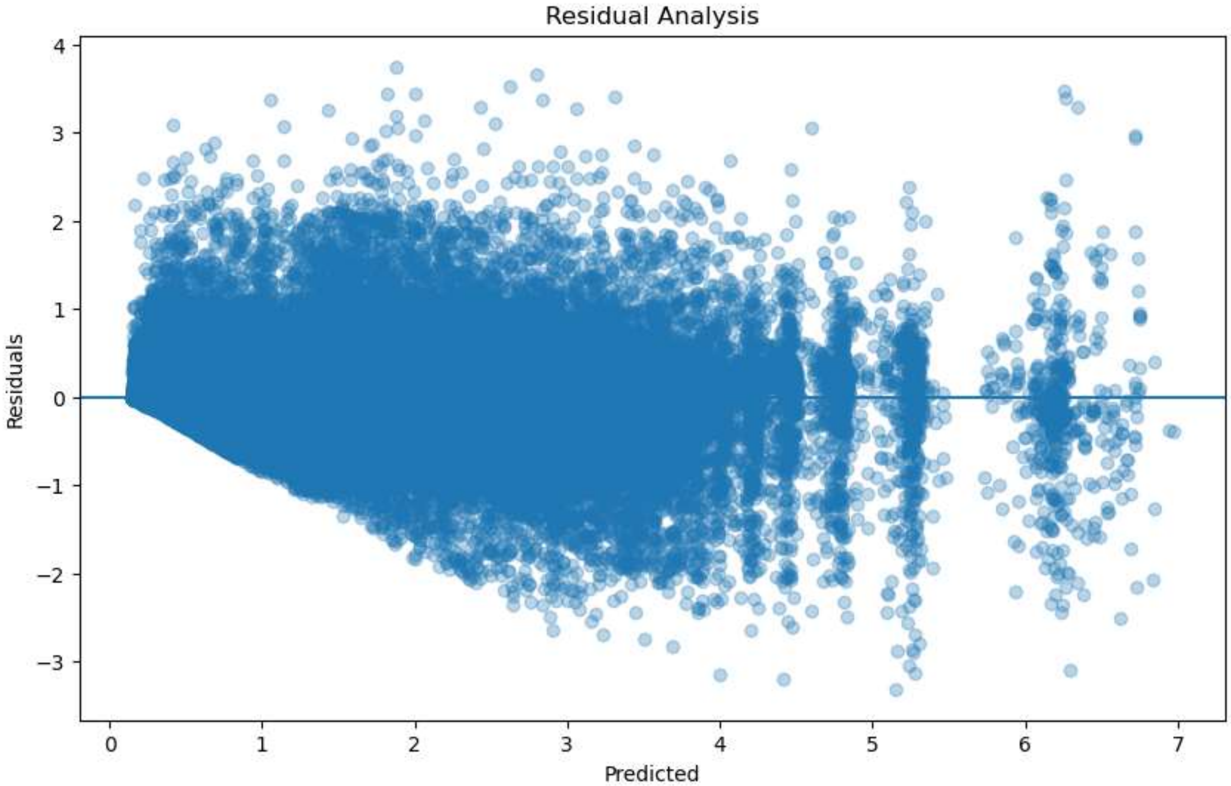
Fig. 2. Illustrates the comparison between actual and predicted electricity demand values for all forecasting models.



The prediction-versus-actual plots (Figure 2) provide additional insight into model behavior. All models generally followed the overall trend of household electricity consumption, particularly during stable periods. However, differences become more noticeable during peak demand fluctuations.

Gradient Boosting and XGBoost closely tracked sudden increases and decreases in energy usage, showing tighter alignment with actual values. Their prediction curves exhibited less deviation during high-variance periods, indicating stronger generalization ability. Linear Regression, while stable, showed slight smoothing effects and occasionally failed to fully capture sharp spikes in demand. Random Forest displayed more visible deviations, particularly during rapid transitions in consumption patterns.

Fig. 3. Presents the residual distributions obtained from the evaluated models.



The residual plots (Figure 3) further confirm these observations. Gradient Boosting and XGBoost produced residuals that were more evenly distributed around zero, suggesting lower bias and better error consistency. Linear Regression showed slightly wider residual spread, while Random Forest exhibited the largest and most irregular residual patterns. Importantly, no strong systematic bias was observed in any model, indicating that the feature engineering process was effective in reducing structural prediction errors.

D. Implications for Household Energy Forecasting

The findings of this study have important implications for residential energy forecasting and demand management. The high predictive performance across all models demonstrates that machine learning techniques, when combined with appropriate feature engineering, can effectively model household electricity consumption patterns.

From a practical perspective, accurate short-term forecasting enables more efficient household energy management and utility planning. Utilities can use such models to better anticipate demand fluctuations, reduce reliance on costly reserve generation, and improve load scheduling efficiency. Even small improvements in forecasting accuracy, as reflected in lower MAE and RMSE values, can translate into significant operational cost savings at scale.

Furthermore, the results highlight the importance of ensemble learning methods, particularly Gradient Boosting and XGBoost, in smart energy systems. Their ability to handle nonlinear relationships and temporal variations makes them highly suitable for integration into intelligent energy management platforms. These models can support demand response programs, dynamic pricing strategies, and real-time grid optimization.

E. Relevance to Smart Grids and Energy Systems

In the context of smart grids, accurate household-level forecasting plays a critical role in maintaining system stability and optimizing energy distribution. Smart grids rely on real-time data analytics to balance supply and demand, especially with increasing penetration of renewable energy sources such as solar and wind.

The strong performance of Gradient Boosting suggests that ensemble-based models can significantly enhance forecasting reliability in such environments. Improved prediction accuracy helps reduce uncertainty in grid operations, enabling better integration of intermittent renewable resources. It also supports more responsive demand-side management strategies, where consumers adjust usage based on predicted or real-time pricing signals.

Additionally, the ability of the models to capture temporal patterns in consumption highlights their potential use in automated energy management systems. These systems could eventually enable households and utilities to make proactive decisions regarding energy usage, storage, and distribution.

These findings further support the work of Ismail et al. [13], who concluded that forecasting performance depends not only on model complexity but also on data quality, feature representation, and forecasting horizon. Accurate household demand prediction therefore remains a fundamental component of sustainable smart-home and smart-grid operation.

F. Limitations of the Study

Despite the promising results, several limitations should be acknowledged. First, the dataset used represents electricity consumption from a single household, which may limit the generalizability of the findings to broader populations. Household energy usage varies significantly across regions, income levels, and climatic conditions.

Second, the models were primarily trained using historical consumption and engineered time-based features. External variables such as temperature, weather conditions, and occupancy behavior were not included, which may have further improved predictive performance.

Third, only four machine learning models were evaluated. Although these models represent a strong baseline comparison, more advanced deep learning architectures such as LSTM or Transformer-based models could potentially capture temporal dependencies more effectively.

Furthermore, the investigation focused on a single forecasting horizon. The relative performance of the evaluated models may differ under ultra-short-term, medium-term, or long-term forecasting scenarios.

G. Summary of Findings

Overall, Gradient Boosting emerged as the most accurate model, closely followed by XGBoost. Linear Regression demonstrated strong baseline performance, while Random Forest showed relatively weaker results in comparison. The results suggest that ensemble learning methods are particularly effective for household energy forecasting, especially when combined with robust feature engineering techniques.

V. CONCLUSION AND FUTURE WORK

This study presented a comparative analysis of machine learning models for short-term household electricity demand forecasting using a real-world smart meter dataset. The primary objective was to evaluate the predictive performance of four widely used algorithms, namely Linear Regression, Random Forest Regression, Gradient Boosting Regression, and Extreme Gradient Boosting (XGBoost), under a unified preprocessing and evaluation framework. By applying consistent feature engineering, including lag variables, rolling statistics, and cyclical time features, the study ensured a fair and robust comparison across all models.

The experimental results demonstrated that all models achieved strong predictive performance, with high R^2 values exceeding 0.93, indicating that the engineered features effectively captured the underlying structure of household electricity consumption. Among the evaluated models, Gradient Boosting achieved the best overall performance, yielding the lowest prediction errors and the highest explanatory power. XGBoost closely followed, while Linear Regression also performed competitively despite its simplicity. Random Forest showed comparatively lower accuracy, although its performance remained acceptable for practical forecasting applications. These results suggest that boosting-based approaches are particularly well-suited for modeling complex household energy consumption patterns.

From a practical perspective, the findings highlight the importance of advanced ensemble learning techniques in improving forecasting accuracy for residential energy systems. More accurate demand predictions can enhance smart grid operations by enabling better load scheduling, reducing reliance on reserve generation, and improving the efficiency of demand-side management strategies. In addition, improved forecasting supports the integration of renewable energy sources by helping balance variable generation with consumer demand, thereby contributing to more stable and sustainable power systems.

The main contribution of this research lies in providing a structured and fair benchmarking framework for evaluating machine learning models in household energy forecasting. Unlike studies that focus on individual algorithms or inconsistent evaluation setups, this work ensures uniform preprocessing, feature engineering, and evaluation conditions, allowing meaningful performance comparisons. The results offer practical insights for both researchers and utility operators in selecting appropriate forecasting models for residential load prediction.

Despite these contributions, the study has certain limitations. First, the analysis was conducted using data from a single household, which may limit the generalizability of the findings to other regions or consumption patterns. Second, only historical consumption and temporal features were considered, while external variables such as weather conditions, occupancy behavior, and electricity pricing were not included. These factors can significantly influence household energy usage and may improve predictive performance if incorporated. Third, the study focused only on classical machine learning models, excluding more advanced deep learning architectures that have shown strong performance in time-series forecasting.

Future research should address these limitations by extending the analysis to multi-household and large-scale datasets to improve model generalization. The integration of exogenous variables such as temperature, humidity, and occupancy data should also be explored to enhance forecasting accuracy. In addition, deep learning approaches such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Transformer-based architectures should be investigated for their ability to capture complex temporal dependencies in energy consumption data. Furthermore, future studies could focus on developing real-time forecasting systems for smart grid applications, enabling dynamic decision-making and adaptive energy management. Hybrid models that combine machine learning with optimization techniques also represent a promising direction for improving forecasting robustness and scalability.

In conclusion, this study demonstrates that carefully engineered temporal features combined with ensemble learning techniques can significantly improve residential electricity demand forecasting accuracy. Among the evaluated algorithms, Gradient Boosting provided the most accurate predictions, achieving the lowest forecasting errors and highest explanatory power. The findings contribute a transparent benchmarking framework for household load forecasting and provide practical guidance for the deployment of machine learning models in smart-grid and energy management applications. As electricity systems become increasingly data-driven and renewable-intensive, accurate demand forecasting will remain a critical component of intelligent energy system operation.

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